Unit-V: Application Layer and Network Security

❖ Functions of Application Layer

The Application Layer, as discussed above, being topmost layer in OSI model, performs several kinds of functions which are requirement in any kind of application or communication process. Following are list of functions which are performed by Application Layer of OSI Model:

Data from User <=> Application layer <=> Data from Presentation Layer

- Application Layer provides a facility by which users can forward several emails and it also provides a storage facility.
- This layer allows users to access, retrieve and manage files in a remote computer.
- It allows users to log on as a remote host.
- This layer provides access to global information about various services.
- This layer provides services which include: e-mail, transferring files, distributing results to the user, directory services, network resources and so on.
- It provides protocols that allow software to send and receive information and present meaningful data to users.
- It handles issues such as network transparency, resource allocation and so on.
- This layer serves as a window for users and application processes to access network services.
- Application Layer is basically not a function, but it performs application layer functions.
- The application layer is actually an abstraction layer that specifies the shared protocols and interface methods used by hosts in a communication network.
- Application Layer helps us to identify communication partners, and synchronizing communication.
- This layer allows users to interact with other software applications.
- In this layer, data is in visual form, which makes users truly understand data rather than remembering or visualize the data in the binary format (0's or 1's).
- This application layer basically interacts with Operating System (OS) and thus further preserves the data in a suitable manner.
- This layer also receives and preserves data from it's previous layer, which is Presentation Layer (which carries in itself the syntax and semantics of the information transmitted).
- The protocols which are used in this application layer depend upon what information users wish to send or receive.
- This application layer, in general, performs host initialization followed by remote login to hosts.

❖ Application Layer Protocol in Computer Network

1. TELNET

Telnet stands for the **TEL**etype **NET**work. It helps in terminal emulation. It allows Telnet clients to access the resources of the Telnet server. It is used for managing files on the Internet. It is used for the initial setup of devices like switches. The telnet command is a command that uses the Telnet protocol to communicate with a remote device or system. The port number of the telnet is 23.

Command

```
telnet [\\RemoteServer]
\\RemoteServer
: Specifies the name of the server
to which you want to connect
```

2. FTP

FTP stands for File Transfer Protocol. It is the protocol that actually lets us transfer files. It can facilitate this between any two machines using it. But FTP is not just a protocol but it is also a program.FTP promotes sharing of files via remote computers with reliable and efficient data transfer. The Port number for FTP is 20 for data and 21 for control.

Command

```
ftp machinename
```

3. TFTP

The Trivial File Transfer Protocol (TFTP) is the stripped-down, stock version of FTP, but it's the protocol of choice if you know exactly what you want and where to find it. It's a technology for transferring files between network devices and is a simplified version of FTP. The Port number for TFTP is 69.

Command

```
tftp [ options... ] [host [port]] [-c command]
```

4. NFS

It stands for a Network File System. It allows remote hosts to mount file systems over a network and interact with those file systems as though they are mounted locally. This enables system administrators to consolidate resources onto centralized servers on the network. The Port number for NFS is 2049.

Command

```
service nfs start
```

5. SMTP

It stands for Simple Mail Transfer Protocol. It is a part of the TCP/IP protocol. Using a process called "store and forward," SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email inbox. The Port number for SMTP is 25.

Command

```
MAIL FROM:<mail@abc.com?
```

6. LPD

It stands for Line Printer Daemon. It is designed for printer sharing. It is the part that receives and processes the request. A "daemon" is a server or agent. The Port number for LPD is 515.

Command

```
lpd [ -d ] [ -l ] [ -D DebugOutputFile]
```

7. X window

It defines a protocol for the writing of graphical user interface–based client/server applications. The idea is to allow a program, called a client, to run on one computer. It is primarily used in networks of interconnected mainframes. Port number for X window starts from 6000 and increases by 1 for each server.

Command

```
Run xdm in runlevel 5
```

8. SNMP

It stands for Simple Network Management Protocol. It gathers data by polling the devices on the network from a management station at fixed or random intervals, requiring them to disclose certain information. It is a way that servers can share information about their current state, and also a channel through which an administrate can modify pre-defined values. The Port number of SNMP is 161(TCP) and 162(UDP).

Command

```
snmpget -mALL -v1 -cpublic snmp_agent_Ip_address sysName.0
```

9. DNS

It stands for Domain Name System. Every time you use a domain name, therefore, a DNS service must translate the name into the corresponding IP address. For example, the domain name www.abc.com might translate to 198.105.232.4. The Port number for DNS is 53.

Command

```
ipconfig /flushdns
```

10. DHCP

It stands for Dynamic Host Configuration Protocol (DHCP). It gives IP addresses to hosts. There is a lot of information a DHCP server can provide to a host when the host is registering for an IP address with the DHCP server. Port number for DHCP is 67, 68.

Command

```
clear ip dhcp binding {address | * }
```

11. HTTP/HTTPS

HTTP stands for Hypertext Transfer Protocol and HTTPS is the more secured version of HTTP, that's why HTTPS stands for Hypertext Transfer Protocol Secure. This protocol is used to access data from the World Wide Web. The Hypertext is the well-organized documentation system that is used to link pages in the text document.

- HTTP is based on the client-server model.
- It uses TCP for establishing connections.
- HTTP is a stateless protocol, which means the server doesn't maintain any information about the previous request from the client.
- HTTP uses port number 80 for establishing the connection.

12. POP

POP stands for Post Office Protocol and the latest version is known as POP3 (Post Office Protocol version 3). This is a simple protocol used by User agents for message retrieval from mail servers.

- POP protocol work with Port number 110.
- It uses TCP for establishing connections.

POP works in dual mode- *Delete mode, Keep Mode*.

In Delete mode, it deletes the message from the mail server once they are downloaded to the local system.

In Keep mode, it doesn't delete the message from the mail server and also facilitates the users to access the mails later from the mail server.

13. IRC

IRC stands for Internet Relay Chat. It is a text-based instant messaging/chatting system. IRC is used for group or one-to-one communication. It also supports file, media, data sharing within the chat. It works upon the client-server model. Where users connect to IRC server or IRC network via some web/ standalone application program.

- It uses TCP or TLS for connection establishment.
- It makes use of port number 6667.

14. MIME

MIME stands for Multipurpose Internet Mail Extension. This protocol is designed to extend the capabilities of the existing Internet email protocol like SMTP. MIME allows non-ASCII data to be sent via SMTP. It allows users to send/receive various kinds of files over the Internet like audio, video, programs, etc. MIME is not a standalone protocol it works in collaboration with other protocols to extend their capabilities.

❖ Architecture of WWW

What is WWW?

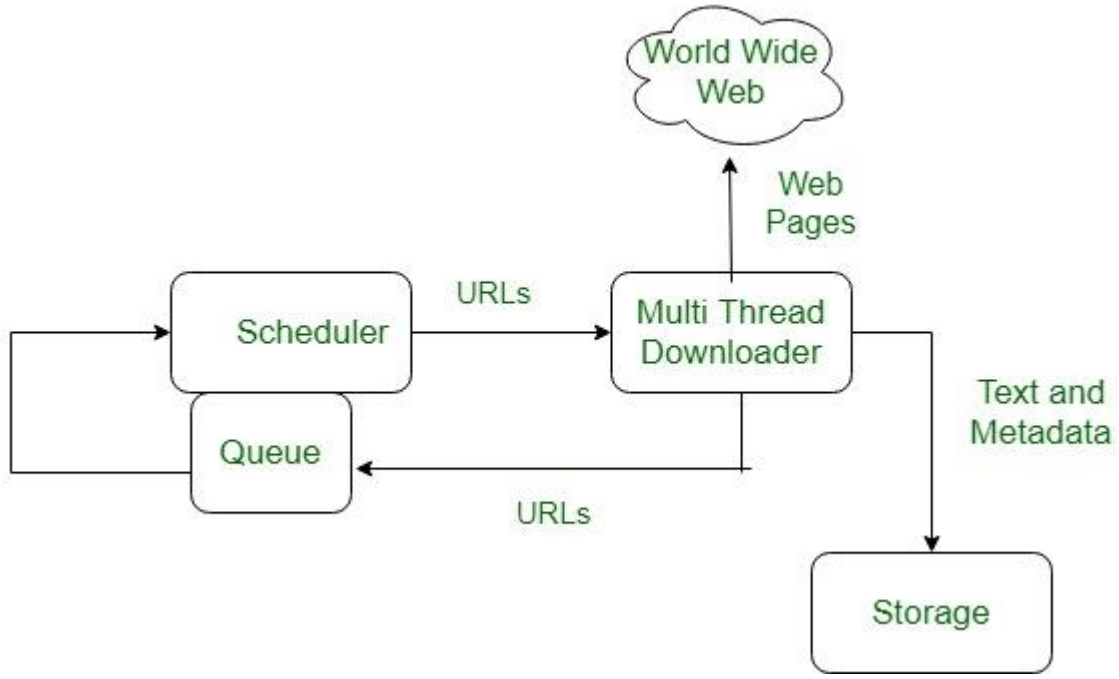
WWW stands for World Wide Web and is commonly known as the Web. The WWW was started by CERN in 1989. WWW is defined as the collection of different websites around the world, containing different information shared via local servers(or computers).

Web pages are linked together using hyperlinks which are HTML-formatted and, also referred to as hypertext, these are the fundamental units of the Internet and are accessed through Hypertext Transfer Protocol(HTTP). Such digital connections, or links, allow users to easily access desired information by connecting relevant pieces of information. The benefit of hypertext is it allows you to pick a word or phrase from the text and click on other sites that have more information about it.

System Architecture

From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google

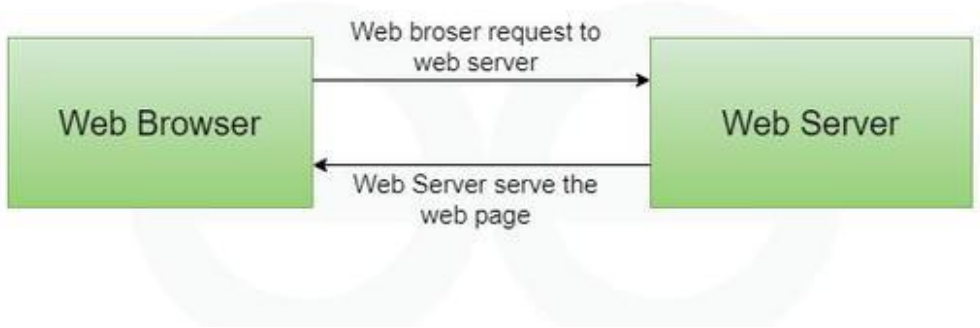
Chrome, etc are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.
The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

Working of WWW

A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal.
The below diagram indicates how the Web operates just like client-server architecture of the internet. When users request web pages or other information, then the web browser of your system request to the server for the information and then the web server provide requested services to web browser back and finally the requested service is utilized by the user who made the request.



World Wide Web

Web browsers can be used for several tasks including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

Features of WWW

- WWW is open source.
- It is a distributed system spread across various websites.
- It is a Hypertext Information System.
- It is Cross-Platform.
- Uses Web Browsers to provide a single interface for many services.
- Dynamic, Interactive and Evolving.

Components of the Web

There are 3 components of the web:

- **Uniform Resource Locator (URL): URL** serves as a system for resources on the web.
- **Hyper Text Transfer Protocol (HTTP): HTTP** specifies communication of browser and server.
- **Hyper Text Markup Language (HTML): HTML** defines the structure, organisation and content of a web page.

What is the Full Form of HTTP?

HTTP stands for "Hypertext Transfer Protocol." It is a set of rules for sharing data on the World Wide Web (WWW). HTTP helps web browsers and servers communicate, allowing people to access and share information over the internet.

Key Points

- **Basic Structure**: HTTP forms the foundation of the web, enabling data communication and file sharing.
- **Web Browsing**: Most websites use HTTP, so when you click on a link or download a file, HTTP is at work.
- **Client-Server Model**: HTTP works on a request-response system. Your browser (client) asks for information, and the website's server responds with the data.
- **Application Layer Protocol**: HTTP operates within the Internet Protocol Suite, managing how data is transmitted and received.

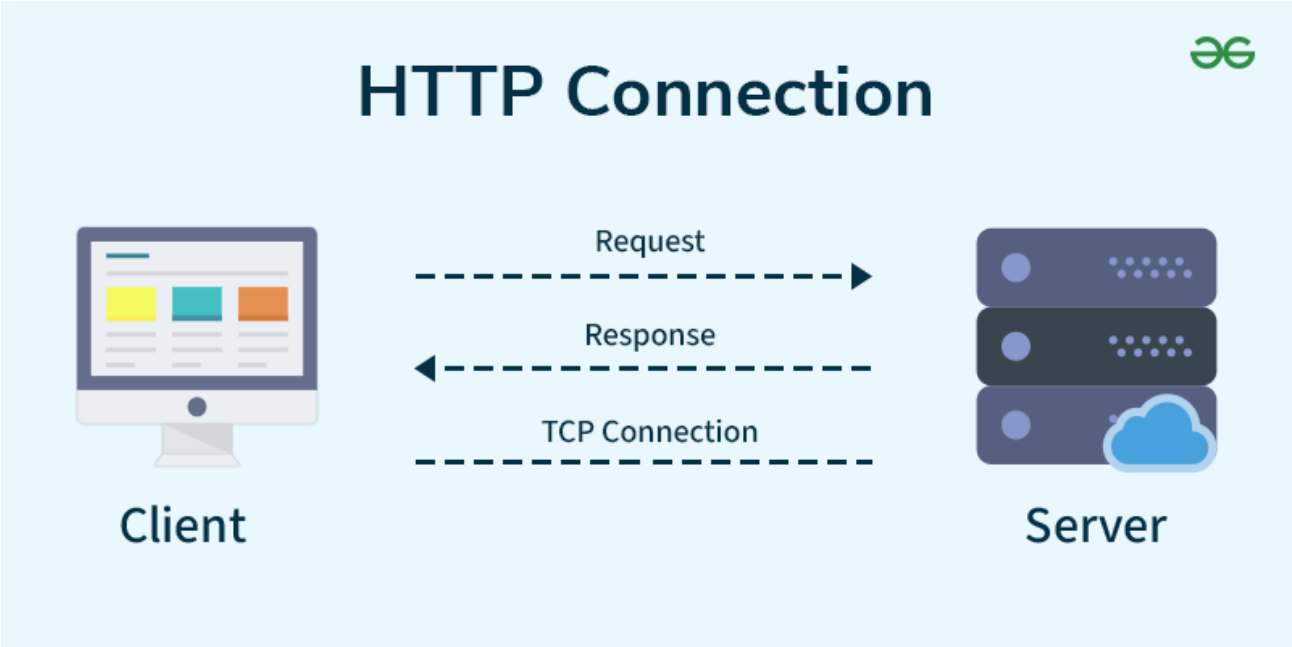
What is HyperText?

The protocol used to transfer hypertext between two computers is known as HyperText Transfer Protocol. HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.

Working of HTTP [HyperText Transfer Protocol]

First of all, whenever we want to open any website we first open a web browser after that we will type the URL of that website (e.g., www.facebook.com). This URL is now sent to the Domain Name Server (DNS). Then DNS first checks records for this URL in their database, and then DNS will return the IP address to the web browser corresponding to this URL. Now the browser is able to send requests to the actual server.

After the server sends data to the client, the connection will be closed. If we want something else from the server we should have to re-establish the connection between the client and the server.



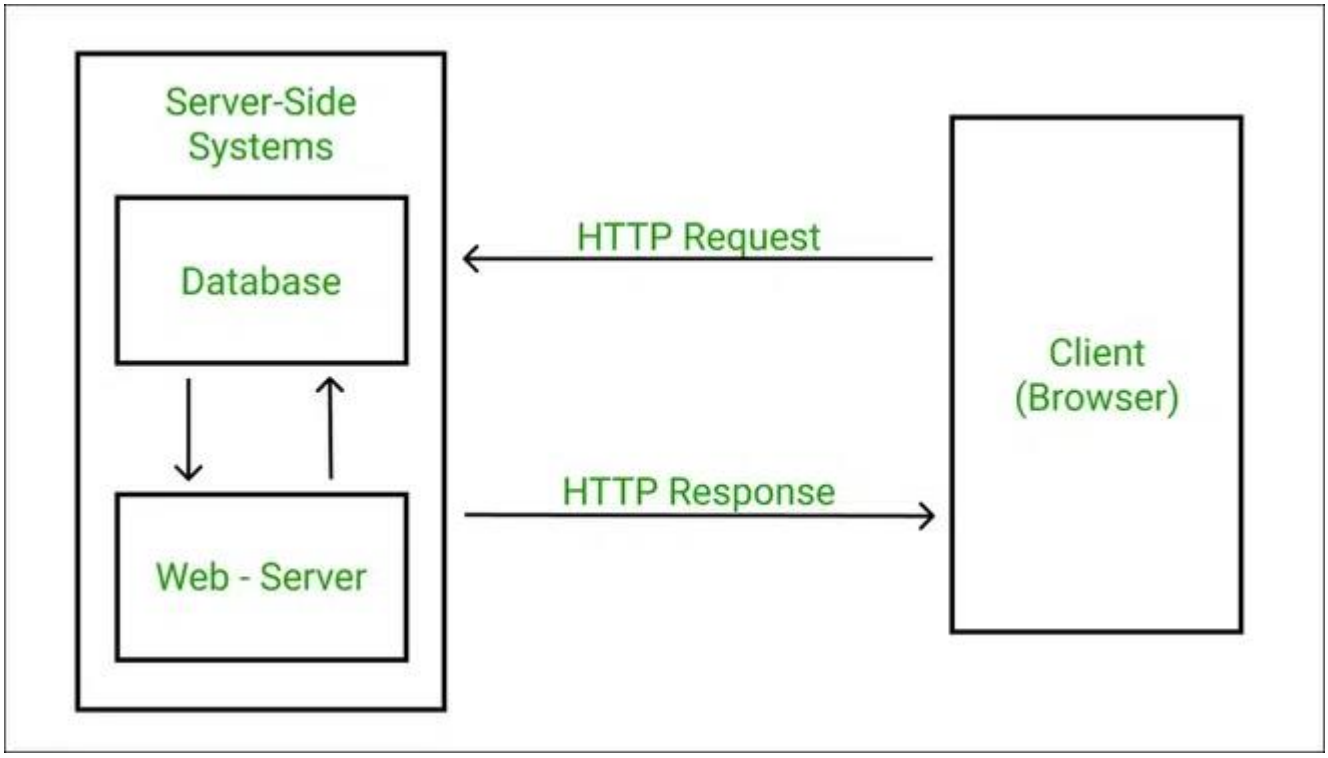
Working off HTTPs

What is an HTTP Request?
HTTP request is simply termed as the information or data that is needed by Internet browsers for loading a website. This is simply known as HTTP Request.
There is some common information that is generally present in all HTTP requests. These are mentioned below.

- HTTP Version
- URL
- HTTP Method
- HTTP Request Headers
- HTTP Body

HTTP Request Headers
HTTP Request Headers generally store information in the form of key-value and must be present in each HTTP Request. The use of this Request Header is to provide core information about the client's information, etc.
HTTP Request Body
HTTP Request Body simply contains the information that has to be transferred. HTTP Request has the information or data to be sent to these browsers.
HTTP Method
HTTP Methods are simply HTTP Verb. In spite of being present so many HTTP Methods, the most common HTTP Methods are HTTP GET and HTTP POST. These two are generally used in HTTP cases. In HTTP GET, the information is received in the form of a website.
What is HTTP Response?
HTTP Response is simply the answer to what a Server gets when the request is raised. There are various things contained in HTTP Response, some of them are listed below.

- HTTP Status Code
- HTTP Headers
- HTTP Body



HTTP Response

HTTP Response Headers
HTTP Response headers are simply like an HTTP Request where it has that work to send some important files and data to the HTTP Response Body.
HTTP Response Body
HTTP Responses are the responses that are received successfully upon the request. Generally, it comes under the requests generated by the web. In most cases, the request is to transfer the HTML data into a webpage.
What is an HTTP Status Code?
HTTP Status Codes are the 3-digit codes that tell the message or simply tell us about the HTTP Request whether it has been completed or not. There are simply 5 types of status codes.

- Informational
- Successful
- Re-directional
- Client-Error
- Server-Error

History of HTTP
Tim Berners Lee and his team at CERN get credit for inventing original HTTP and associated technologies.

- **HTTP version 0.9:** This was the first version of HTTP which was introduced in 1991.
- **HTTP version 1.0:** In 1996, RFC 1945 (Request For Comments) was introduced in HTTP version 1.0.
- **HTTP version 1.1:** In January 1997, RFC 2068 was introduced in HTTP version 1.1. Improvements and updates to the HTTP version 1.1 standard were released under RFC 2616 in June 1999.
- **HTTP version 2.0:** The HTTP version 2.0 specification was published as RFC 7540 on May 14, 2015.

- **HTTP version 3.0:** HTTP version 3.0 is based on the previous RFC draft. It is renamed as Hyper-Text Transfer Protocol QUIC which is a transport layer network protocol developed by Google.

Characteristics of HTTP

HTTP is IP based communication protocol that is used to deliver data from server to client or vice-versa.

- The server processes a request, which is raised by the client, and also server and client know each other only during the current bid and response period.
- Any type of content can be exchanged as long as the server and client are compatible with it.
- Once data is exchanged, servers and clients are no longer connected.
- It is a request and response protocol based on client and server requirements.
- It is a connection-less protocol because after the connection is closed, the server does not remember anything about the client and the client does not remember anything about the server.
- It is a stateless protocol because both client and server do not expect anything from each other but they are still able to communicate.

Advantages of HTTP

- Memory usage and CPU usage are low because of fewer simultaneous connections.
- Since there are few TCP connections hence network congestion is less.
- Since handshaking is done at the initial connection stage, then latency is reduced because there is no further need for handshaking for subsequent requests.
- The error can be reported without closing the connection.
- HTTP allows HTTP pipe-lining of requests or responses.

Disadvantages of HTTP

- HTTP requires high power to establish communication and transfer data.
- HTTP is less secure because it does not use any encryption method like HTTPS and uses TLS to encrypt regular HTTP requests and responses.
- HTTP is not optimized for cellular phones and it is too gabby.
- HTTP does not offer a genuine exchange of data because it is less secure.
- The client does not close the connection until it receives complete data from the server; hence, the server needs to wait for data completion and cannot be available for other clients during this time.

Cryptography Introduction

Cryptography is the study and practice of techniques for secure communication in the presence of third parties called adversaries. It deals with developing and analysing protocols that prevents malicious third parties from retrieving information being shared between two entities thereby following the various aspects of information security. Secure Communication refers to the scenario where the message or data shared between two parties can't be accessed by an adversary. In Cryptography, an Adversary is a malicious entity, which aims to retrieve precious information or data thereby undermining the principles of information security. Data Confidentiality, Data Integrity, Authentication and Non-repudiation are core principles of modern-day cryptography.

1. **Confidentiality** refers to certain rules and guidelines usually executed under confidentiality agreements which ensure that the information is restricted to certain people or places.
2. **Data integrity** refers to maintaining and making sure that the data stays accurate and consistent over its entire life cycle.
3. **Authentication** is the process of making sure that the piece of data being claimed by the user belongs to it.
4. **Non-repudiation** refers to the ability to make sure that a person or a party associated with a contract or a communication cannot deny the authenticity of their signature over their document or the sending of a message.

Consider two parties Alice and Bob. Now, Alice wants to send a message m to Bob over a secure channel. So, what happens is as follows. The sender's message or sometimes called the Plaintext, is converted into an unreadable form using a Key k. The resultant text obtained is called the Ciphertext. This process is known as Encryption. At the time of received, the Ciphertext is converted back into the plaintext using the same Key k, so that it can be read by the receiver. This process is known as Decryption.

```
Alice (Sender)      Bob (Receiver)

C = E (m, k)  ---->    m = D (C, k)
```

Here, C refers to the Ciphertext while E and D are the Encryption and Decryption algorithms respectively. Let's consider the case of Caesar Cipher or Shift Cipher as an example. As the name suggests, in Caesar's Cipher each character in a word is replaced by another character under some defined rules. Thus, if A is replaced by D, B by E and so on. Then, each character in the word would be shifted by a position of 3. For example:

```
Plaintext : Geeksforgeeks

Ciphertext : Jhhnvirujhhnv
```

Note: *Even if the adversary knows that the cipher is based on Caesar's Cipher, it cannot predict the plaintext as it doesn't have the key in this case which is to shift the characters back by three places. Refer to Introduction to Crypto-terminologies.*

Types of Cryptography:

There are several types of cryptography, each with its own unique features and applications. Some of the most common types of cryptography include:

1. Symmetric-key cryptography: This type of cryptography involves the use of a single key to encrypt and decrypt data. Both the sender and receiver use the same key, which must be kept secret to maintain the security of the communication.

2. Asymmetric-key cryptography: Asymmetric-key cryptography, also known as public-key cryptography, uses a pair of keys – a public key and a private key – to encrypt and decrypt data. The public key is available to anyone, while the private key is kept secret by the owner.
Hash functions: A hash function is a mathematical algorithm that converts data of any size into a fixed-size output. Hash functions are often used to verify the integrity of data and ensure that it has not been tampered with.

Applications of Cryptography:

Cryptography has a wide range of applications in modern-day communication, including:

- **Secure online transactions:** Cryptography is used to secure online transactions, such as online banking and e-commerce, by encrypting sensitive data and protecting it from unauthorized access.
- **Digital signatures:** Digital signatures are used to verify the authenticity and integrity of digital documents and ensure that they have not been tampered with.
- **Password protection:** Passwords are often encrypted using cryptographic algorithms to protect them from being stolen or intercepted.

Military and intelligence applications: Cryptography is widely used in military and intelligence applications to protect classified information and communications.

Challenges of Cryptography:

While cryptography is a powerful tool for securing information, it also presents several challenges, including:

- **Key management**: Cryptography relies on the use of keys, which must be managed carefully to maintain the security of the communication.
- **Quantum computing:** The development of quantum computing poses a potential threat to current cryptographic algorithms, which may become vulnerable to attacks.
- **Human error:** Cryptography is only as strong as its weakest link, and human error can easily compromise the security of a communication.

What is Network Security?
Any action intended to safeguard the integrity and usefulness of your data and network is known as network security. In other words, Network security is defined as the activity created to protect the integrity of your network and data.
Network security is the practice of protecting a computer network from unauthorized access, misuse, or attacks. It involves using tools, technologies, and policies to ensure that data traveling over the network is safe and secure, keeping sensitive information away from hackers and other threats.

Network Security

How Does Network Security Work?

Network security uses several layers of protection, both at the edge of the network and within it. Each layer has rules and controls that determine who can access network resources. People who are allowed access can use the network safely, but those who try to harm it with attacks or other threats are stopped from doing so.

The basic principle of network security is protecting huge stored data and networks in layers that ensure the bedding of rules and regulations that have to be acknowledged before performing any activity on the data. These levels are:

- **Physical Network Security:** This is the most basic level that includes protecting the data and network through unauthorized personnel from acquiring control over the confidentiality of the network. The same can be achieved by using devices like biometric systems.
- **Technical Network Security:** It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.
- **Administrative Network Security:** This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

Types of Network Security

There are several types of network security through which we can make our network more secure, Your network and data are shielded from breaches, invasions, and other dangers by network security. Here below are some important types of network security:

Email Security

Email Security is defined as the process designed to protect the Email Account and its contents safe from unauthorized access. For Example, you generally see, fraud emails are automatically sent to the Spam folder. because most email service providers have built-in features to protect the content. The most common danger vector for a security compromise is email gateways. Hackers create intricate phishing campaigns using recipients' personal information and social engineering techniques to trick them and direct them to malicious websites. To stop critical data from being lost, an email security programme restricts outgoing messages and stops incoming threats.

Network Segmentation

Network traffic is divided into several categories by software-defined segmentation, which also facilitates the enforcement of security regulations. Ideally, endpoint identity—rather than just IP addresses—is the basis for the classifications. To ensure that the appropriate amount of access is granted to the appropriate individuals and that suspicious devices are controlled and remediated, access permissions can be assigned based on role, location, and other factors.

Access Control

Your network should not be accessible to every user. You need to identify every user and every device in order to keep out any attackers. You can then put your security policies into effect. Noncompliant endpoint devices might either have their access restricted or blocked. Network access control (NAC) is this process.

Sandboxing

Sandboxing is a cybersecurity technique in which files are opened or code is performed on a host computer that simulates end-user operating environments in a secure, isolated environment. To keep threats off the network, sandboxing watches the code or files as they are opened and searches for harmful activity.

Cloud Network Security

This is very vulnerable to the malpractices that few unauthorized dealers might pertain to. This data must be protected and it should be ensured that this protection is not jeopardized by anything. Many businesses embrace SaaS applications for providing some of their employees the allowance of accessing the data stored in the cloud. This type of security ensures creating gaps in the visibility of the data.

Workloads and applications are no longer solely housed in a nearby data centre on-site. More adaptability and creativity are needed to protect the modern data centre as application workloads move to the cloud.

Web Security

A online security solution will restrict access to harmful websites, stop web-based risks, and manage staff internet usage. Your web gateway will be safeguarded both locally and in the cloud. "Web security" also include the precautions you take to safeguard your personal website.

Intrusion Prevention System(IPS)

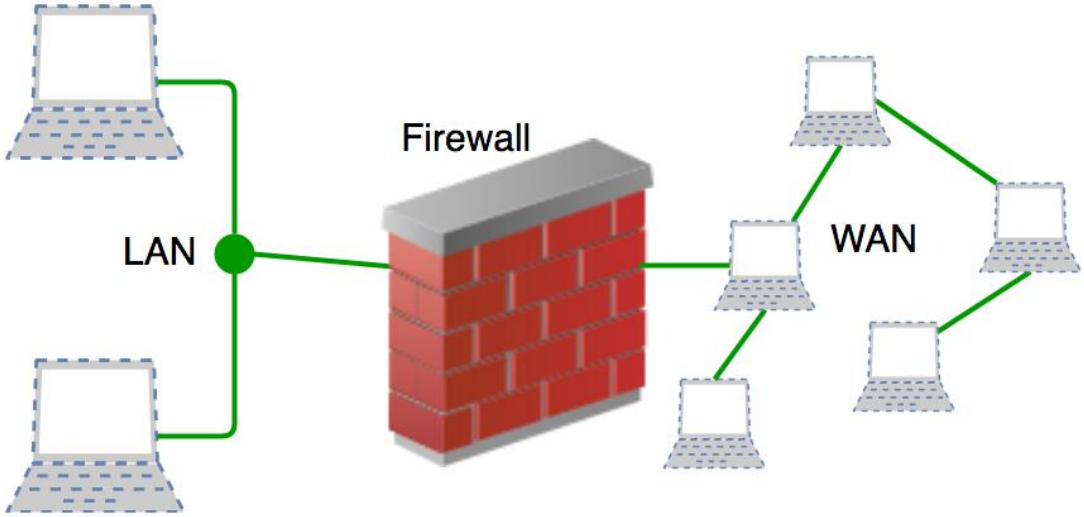
An intrusion Prevention System is also known as Intrusion Detection and Prevention System. It is a network security application that monitors network or system activities for malicious activity. The major functions of intrusion prevention systems are to identify malicious activity, collect information about this activity, report it, and attempt to block or stop it.

Antivirus and Anti-malware Software

This type of network security ensures that any malicious software does not enter the network and jeopardize the security of the data. Malicious software like Viruses, Trojans, and Worms is handled by the same. This ensures that not only the entry of the malware is protected but also that the system is well-equipped to fight once it has entered.

Firewalls Security

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules accepts, rejects, or drops that specific traffic. Before Firewalls, network security was performed by Access Control Lists (ACLs) residing on routers.



Application Security

Application security denotes the security precautionary measures utilized at the application level to prevent the stealing or capturing of data or code inside the application. It also includes the security measurements made during the advancement and design of applications, as well as techniques and methods for protecting the applications whenever.

Wireless Security

Wireless networks are less secure than wired ones. If not properly secured, setting up a wireless LAN can be like having Ethernet ports available everywhere, even in places like parking lots. To prevent attacks and keep your wireless network safe, you need dedicated products designed to protect it from exploits and unauthorized access.

Web Security

A web security solution manages how your staff uses the internet, blocks threats from websites, and stops access to harmful sites. It safeguards your web gateway either onsite or in the cloud. Additionally, "web security" involves measures taken to protect your own website from potential attacks and vulnerabilities.

Mobile Device Security

Cybercriminals are focusing more on mobile devices and apps. In the next three years, about 90 percent of IT organizations might allow corporate applications on personal mobile devices. It's crucial to control which devices can connect to your network and set up their connections securely to protect network traffic from unauthorized access.

Industrial Network Security

As industries digitize their operations, the closer integration of IT, cloud services, and industrial networks exposes Industrial Control Systems (ICS) to cyber threats. To safeguard against these risks, it's crucial to have complete visibility into your Operational Technology (OT) security status. This involves segmenting the industrial network and providing detailed information about OT devices and their behaviors to IT security tools. This approach helps in effectively monitoring and protecting critical industrial systems from potential cyber attacks.

VPN Security

A virtual private network (VPN) encrypts the connection between a device and a network, usually over the internet. A remote-access VPN commonly uses IPsec or Secure Sockets Layer (SSL) to verify and secure the communication between the device and the network. This encryption ensures that data transmitted between the device and the network remains private and secure from unauthorized access.

Benefits of Network Security

Network Security has several benefits, some of which are mentioned below:

- Network Security helps in protecting clients' information and data which ensures reliable access and helps in protecting the data from cyber threats.
- Network Security protects the organization from heavy losses that may have occurred from data loss or any security incident.
- It overall protects the reputation of the organization as it protects the data and confidential items.

Advantages of Network Security

- **Protection from Unauthorized Access**: Network security measures such as firewalls and authentication systems prevent unauthorized users from accessing sensitive information or disrupting network operations.
- **Data Confidentiality**: Encryption technologies ensure that data transmitted over the network remains confidential and cannot be intercepted by unauthorized parties.
- **Prevention of Malware and Viruses**: Network security solutions like antivirus software and intrusion detection systems (IDS) detect and block malware, viruses, and other malicious threats before they can infect systems.
- **Secure Remote Access**: Virtual private networks (VPNs) and other secure remote access methods enable employees to work remotely without compromising the security of the organization's network and data.

Disadvantages of Network Security

- **Complexity and Management Overhead**: Implementing and managing network security measures such as firewalls, encryption, and intrusion detection systems (IDS) can be complex and require specialized knowledge and resources.
- **Cost**: Effective network security often requires investment in hardware, software, and skilled personnel, which can be expensive for organizations, especially smaller ones.
- **Privacy Concerns**: Some network security measures, such as deep packet inspection and monitoring, may raise privacy concerns among users and stakeholders, requiring careful balancing of security needs with individual privacy rights.

❖ **What is Symmetric Key Encryption?**

Encryption is a process to change the form of any message to protect it from reading by anyone. In Symmetric-key encryption the message is encrypted by using a key and the same key is used to decrypt the message which makes it easy to use but less secure. It also requires a safe method to transfer the key from one party to another.

What is Asymmetric Key Encryption?

Asymmetric key encryption is one of the most common cryptographic methods that involve using a single key and its pendent, where one key is used to encrypt data and the second one is used to decrypt an encrypted text. The second key is kept highly secret, while the first one which is called a public key can be freely distributed among the service's users.

Difference Between Symmetric and Asymmetric Key Encryption

Symmetric Key Encryption	Asymmetric Key Encryption
It only requires a single key for both encryption and decryption.	It requires two keys, a public key and a private key, one to encrypt and the other to decrypt.

Symmetric Key Encryption	Asymmetric Key Encryption
The size of ciphertext is the same or smaller than the original plaintext.	The size of ciphertext is the same or larger than the original plaintext.
The encryption process is very fast.	The encryption process is slow.
It is used when a large amount of data needs to be transferred. It only provides confidentiality.	It is used to transfer small amount of data. It provides confidentiality, authenticity, and non-repudiation.
The length of key used is 128 or 256 bits	The length of key used is 2048 or higher
In symmetric key encryption, resource utilization is low compared to asymmetric key encryption.	In asymmetric key encryption, resource utilization is high.
It is efficient as it is used for handling large amount of data. Security is lower as only one key is used for both encryption and decryption purposes.	It is comparatively less efficient as it can handle a small amount of data. Security is higher as two keys are used, one for encryption and the other for decryption.
The Mathematical Representation is as follows- P = D (K, E(K, P)) where K –> encryption and decryption key P –> plain text D –> Decryption E(K, P) –> Encryption of plain text using K **Examples:** 3DES, AES, DES and RC4	The Mathematical Representation is as follows- P = D(Kd, E (Ke,P)) where Ke –> encryption key Kd –> decryption key D –> Decryption E(Ke, P) –> Encryption of plain text using encryption key Ke. P –> plain text **Examples:** Diffie-Hellman, ECC, El Gamal, DSA and RSA

- **Digital Signature**

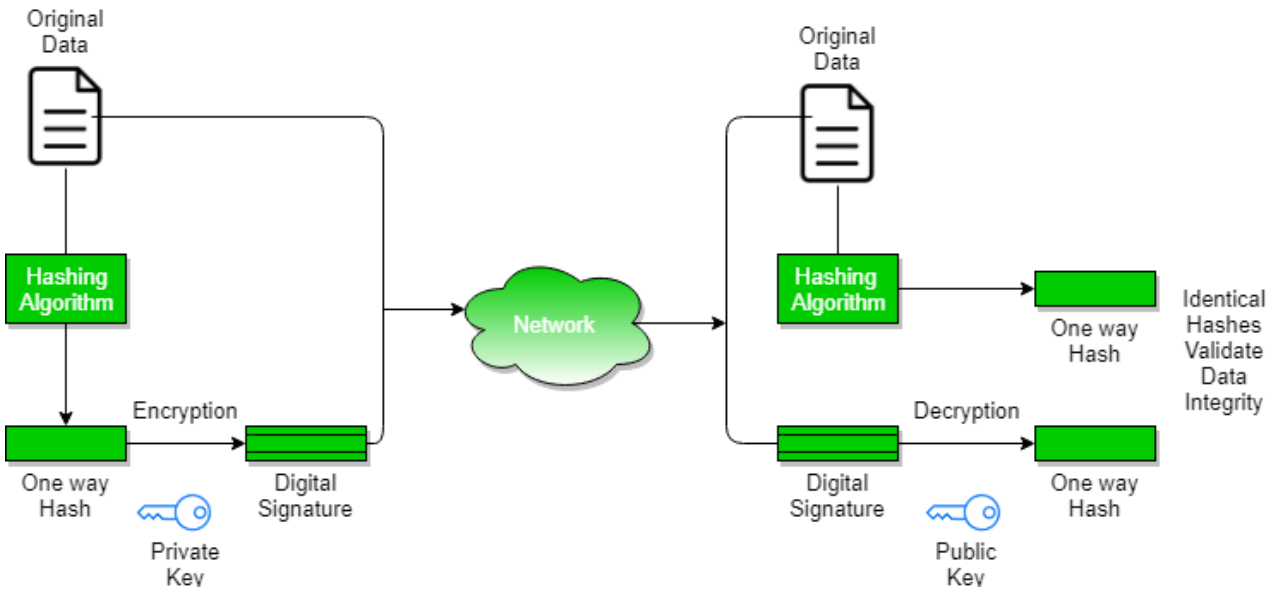
A digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software, or digital document.

1. **Key Generation Algorithms**: Digital signature is electronic signatures, which assure that the message was sent by a particular sender. While performing digital transactions authenticity and integrity should be assured, otherwise, the data can be altered or someone can also act as if he was the sender and expect a reply.
2. **Signing Algorithms**: To create a digital signature, signing algorithms like email programs create a one-way hash of the electronic data which is to be signed. The signing algorithm then encrypts the hash value using the private key (signature key). This encrypted hash along with other information like the hashing algorithm is the digital signature. This digital signature is appended with the data and sent to the verifier. The reason for encrypting the hash instead of the entire message or document is that a hash function converts any arbitrary input into a much shorter fixed-length value. This saves time as now instead of signing a long message a shorter hash value has to be signed and moreover hashing is much faster than signing.
3. **Signature Verification Algorithms** : Verifier receives Digital Signature along with the data. It then uses Verification algorithm to process on the digital signature and the public key (verification key) and generates some value. It also applies the same hash function on the received data and generates a hash value. If they both are equal, then the digital signature is valid else it is invalid.

The steps followed in creating digital signature are :

1. Message digest is computed by applying hash function on the message and then message digest is encrypted using private key of sender to form the digital signature. (digital signature = encryption (private key of sender, message digest) and message digest = message digest algorithm(message)).
2. Digital signature is then transmitted with the message.(message + digital signature is transmitted)
3. Receiver decrypts the digital signature using the public key of sender.(This assures authenticity, as only sender has his private key so only sender can encrypt using his private key which can thus be decrypted by sender's public key).
4. The receiver now has the message digest.
5. The receiver can compute the message digest from the message (actual message is sent with the digital signature).
6. The message digest computed by receiver and the message digest (got by decryption on digital signature) need to be same for ensuring integrity.

Message digest is computed using one-way hash function, i.e. a hash function in which computation of hash value of a message is easy but computation of the message from hash value of the message is very difficult.



Assurances about digital signatures

The definitions and words that follow illustrate the kind of assurances that digital signatures offer.

1. **Authenticity**: The identity of the signer is verified.
2. **Integration:** Since the content was digitally signed, it hasn't been altered or interfered with.
3. **Non-repudiation:** demonstrates the source of the signed content to all parties. The act of a signer denying any affiliation with the signed material is known as repudiation.
4. **Notarization:** Under some conditions, a signature in a Microsoft Word, Microsoft Excel, or Microsoft PowerPoint document that has been time-stamped by a secure time-stamp server is equivalent to a notarization.

Benefits of Digital Signatures

- **Legal documents and contracts:** Digital signatures are legally binding. This makes them ideal for any legal document that requires a signature authenticated by one or more parties and guarantees that the record has not been altered.
- **Sales contracts:** Digital signing of contracts and sales contracts authenticates the identity of the seller and the buyer, and both parties can be sure that the signatures are legally binding and that the terms of the agreement have not been changed.
- **Financial Documents:** Finance departments digitally sign invoices so customers can trust that the payment request is from the right seller, not from a bad actor trying to trick the buyer into sending payments to a fraudulent account.
- **Health Data:** In the healthcare industry, privacy is paramount for both patient records and research data. Digital signatures ensure that this confidential information was not modified when it was transmitted between the consenting parties.

****